

Some Special functions and Dwork's lemma (Lunch Seminar)

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Based on Section 2 of Chapter 4 of [Kob96]. In this lecture, we will continue to talk about analytic functions, including studying their zeros via Newton polygons.

Claim 1. *There exists a unique function $\log_p: \mathbb{C}_p^\times \rightarrow \mathbb{C}_p$ such that:*

1. $\log_p$ agrees with the previous definition on $B_1(1)$: $\log_p x = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{n}$.
2. $\forall x, y \in \mathbb{C}_p^\times: \log_p(xy) = \log_p(x) + \log_p(y)$.
3. $\log_p p = 0$

Lemma 2. *Every $x \in \mathbb{C}_p^\times$ can be written as $x = p^r \cdot \zeta \cdot x_1$ where $r \in \mathbb{Q}$, ζ is a root of unity and $|x_1 - 1|_p < 1$*

Proof. Firstly, note that $|x|_p = p^r$ for some $r \in \mathbb{Q}$: Take a sequence $(x_n)_{n=1}^{\infty}$ in $\overline{\mathbb{Q}_p}$ converging to x . If $|x_n|_p$ is not eventually constant then we can find arbitrarily large $n, m \in \mathbb{N}$ such that $|x_n|_p \neq |x_m|_p$ and so $|x_n - x_m|_p = \max\{|x_n|_p, |x_m|_p\}$ but it is a Cauchy sequence so for every $\varepsilon > 0$ we get $\max\{|x_n|_p, |x_m|_p\} < \varepsilon$ for some large enough n, m meaning there is a subsequence of $(x_n)_{n=1}^{\infty}$ converging to 0 and we get a contradiction. So $(|x_n|_p)_{n=1}^{\infty}$ is eventually constant with values in $p^{\mathbb{Q}}$ and $|x|_p$ is its limit.

Now we choose some $p^r \in \overline{\mathbb{Q}_p}$ (choice because we take roots) and write $x = p^r \cdot x'$ (so $|x'|_p = 1$). Denote $A = \{a \in \mathbb{C}_p \mid |a|_p \leq 1\}$, $M = \{a \in \mathbb{C}_p \mid |a|_p < 1\}$. Take $x'' \in \overline{\mathbb{Q}_p}$ such that $|x' - x''|_p < 1$ (exists from density), this implies that $|x''|_p = 1$ and $x' + M = x'' + M$ in A/M , which is some large extension of $\mathbb{F}_p$. Hence this coset is algebraic over $\mathbb{F}_p$ (recall x'' is algebraic and a root of a monic polynomial over $\mathbb{Z}_p$, then reduce that polynomial), so it is in some $\mathbb{F}_{p^f}$ and there is a root of unity ζ such that $\zeta^{-1}x'' = 1$ in A/M which means that $\zeta^{-1}x' + M = \zeta^{-1}x'' + M = 1 + M \Rightarrow \zeta^{-1}x' - 1 \in M$ and we finish by taking $x_1 := \zeta^{-1}x'$. $\square$

sketch of claim's proof. For every $x \in \mathbb{C}_p^\times$ write $x = p^r \cdot \zeta \cdot x_1$ as in lemma. From properties 2,3 we have $\log_p(p^r) = \log_p(\zeta) = 0$ so the only possible definition is $\log_p(x) := \log_p(x_1)$ and now we need to check that this definition is well defined and satisfies all the properties (which we won't do). $\square$

Claim 3.

1. $\log_p$ is locally analytic, meaning around every point there is a power series expansion with positive radius of convergence.
2. $\frac{d}{dx} \log_p(x) = \frac{1}{x}$

Proof. 1. For every $x_0 \in \mathbb{C}_p^\times$, denote $r := |x_0|_p > 0$. For every $x \in B_r(x_0)$ we have $|\frac{x}{x_0} - 1|_p < 1$, so:

$$\log_p(x) = \log_p(x_0) + \log\left(\frac{x}{x_0}\right) = \log_p(x_0) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(\frac{x}{x_0} - 1)^n}{n} = \log_p(x_0) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x - x_0)^n}{x_0^n n}$$

2. Differentiating term by term the power series expansion above, we get

$$\begin{aligned}\frac{d}{dx} \log_p(x) &= \sum_{n=1}^{\infty} \frac{n(-1)^{n+1}(x-x_0)^{n-1}}{x_0^n n} = x_0^{-1} \sum_{n=0}^{\infty} \left(\frac{x_0-x}{x_0} \right)^n = x_0^{-1} \sum_{n=0}^{\infty} \left(1 - \frac{x}{x_0} \right)^n = \\ &= x_0^{-1} \frac{1}{1 - (1 - \frac{x}{x_0})} = \frac{1}{x}\end{aligned}$$

□

Now we turn to the exponent function.

Reminder 4. Denote by $\mu: \mathbb{N} \rightarrow \mathbb{Z}$ the Mobius function $\mu(n) = \begin{cases} 0 & k^2 \mid n \text{ for some } k > 1 \\ (-1)^k & n = p_1 \cdot \dots \cdot p_k \text{ distinct primes} \end{cases}$.

Then we have the identity

$$\sum_{0 < d \mid n} \mu(d) = \begin{cases} 1 & n = 1 \\ 0 & n \neq 1 \end{cases}$$

Claim 5. In $\mathbb{Q}[[x]]$ we have the following formal identity $e^x = \prod_{n=1}^{\infty} (1 - x^n)^{-\mu(n)/n}$, where the right hand side is genuinely an element in $\mathbb{Q}[[x]]$ because $(1+x)^a = \sum_{n=0}^{\infty} \binom{a}{n} x^n$ and in that infinite product every coefficient is actually a product of finitely many things.

Proof. Take log of the right hand side:

$$\begin{aligned}\log\left(\prod_{n=1}^{\infty} (1 - X^n)^{-\mu(n)/n}\right) &= - \sum_{n=1}^{\infty} \frac{\mu(n)}{n} \log(1 - x^n) = \sum_{n=1}^{\infty} \frac{\mu(n)}{n} \sum_{m=1}^{\infty} \frac{x^{nm}}{m} = \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \mu(n) \frac{x^{nm}}{nm} = \sum_{k=1}^{\infty} \left(\sum_{0 < d \mid k} \mu(d) \right) \frac{x^k}{k} = x\end{aligned}$$

□

We know that e^x converges only on a disc of radius $p^{-1/(p-1)}$, so if we would want to define a function similar to exponentiation that converges on a larger disc, we may look at the right hand side and check which terms in the product are the problematic ones. it turns out that those are n 's which are multiples of p , for example if $n = p$ then

$$(1 - x^p)^{-\mu(p)/p} = (1 - x^p)^{1/p} = \sum_{k=1}^{\infty} \binom{1/p}{k} (-x^p)^k$$

Definition 6. The Artin-Hasse exponential is defined by

$$E_p(x) = \prod_{n=1, p \nmid n}^{\infty} (1 - x^n)^{-\mu(n)/n}$$

In fact, we can find a nicer expression for E_p

Claim 7.

$$E_p(x) = e^{x + \frac{x^p}{p} + \frac{x^{p^2}}{p^2} + \dots}$$

Proof. Taking log, we get

$$\log(E_p(x)) = \dots = \sum_{k=1}^{\infty} \left(\underbrace{\sum_{\substack{0 < d|k, \\ p \nmid d}} \mu(d)}_{(*)} \right) \frac{x^k}{k}$$

Note that $(*) = 1$ if k is a power of p and otherwise 0, by using the previous identity for $\frac{n}{p^{\text{ord}_p(n)}}$. So we get

$$\log(E_p(x)) = \sum_{k=0}^{\infty} \frac{x^{p^k}}{p^k}$$

□

Lemma 8 (Dwork's lemma). *Let $f(x) \in 1 + x\mathbb{Q}_p[[x]]$. Then $f \in 1 + x\mathbb{Z}_p[[x]]$ if and only if $\frac{f(x^p)}{f(x)^p} \in 1 + px\mathbb{Z}_p[[x]]$*

we skip the proof because of lack of time.

Remark. The quotient $\frac{f(x^p)}{f(x)^p}$ essentially measures "how close to commuting are f and raising to p -th power?", which is apparently important in p -adic analysis. The lemma says that the coefficients of f are p -adic integers if and only if f and raising to the p -th power "commute mod p ".

In addition, this lemma works the same for multivariate power series (and we will use this fact).

Corollary 9. $E_p \in 1 + x\mathbb{Z}_p[[x]]$ and so it converges in $B_0(1)$

Proof. using the previous claim, we get

$$\frac{E_p(x^p)}{E_p(x)^p} = \frac{e^{x^p + x^{p^2}/p + x^{p^3}/p^2 + \dots}}{e^{px + x^p + x^{p^2}/p + x^{p^3}/p^2 + \dots}} = e^{-px} \in 1 + px\mathbb{Z}_p[[x]]$$

and then by the lemma $E_p \in 1 + x\mathbb{Z}_p[[x]]$

□

We now turn to a function that will be used in Dwork's proof of the rationality of the zeta function.

Definition 10. Define $F \in \mathbb{Q}[[x, y]]$ by

$$F(x, y) = (1 + y)^x \cdot (1 + y^p)^{(x^p - x)/p} \cdot (1 + y^{p^2})^{(x^{p^2} - x^p)/p^2} \cdot \dots$$

and again this is genuinely in $\mathbb{Q}[[x, y]]$ as we open the definition of the generalized binomial and every coefficient in the infinite product is defined by a finite product.

Claim 11. $F \in 1 + x\mathbb{Z}_p[[x, y]] + y\mathbb{Z}_p[[x, y]]$

Proof. We will use Dwork's lemma generalized to 2 variables.

$$\frac{F(x^p, y^p)}{F(x, y)^p} = \frac{(1 + y^p)^{x^p} \cdot (1 + y^{p^2})^{(x^{p^2} - x^p)/p} \cdot (1 + y^{p^3})^{(x^{p^3} - x^{p^2})/p^2} \cdot \dots}{(1 + y)^{px} \cdot (1 + y^p)^{(x^p - x)} \cdot (1 + y^{p^2})^{(x^{p^2} - x^p)/p} \cdot \dots} = \frac{(1 + y^p)^x}{(1 + y)^{px}}$$

Note that $1 + y \in 1 + y\mathbb{Z}_p[[y]]$ so by the lemma there is $G(y) \in \mathbb{Z}_p[[y]]$ such that

$$\begin{aligned} \frac{1 + y^p}{(1 + y)^p} &= 1 + pyG(y) \\ \Rightarrow \frac{(1 + y^p)^x}{(1 + y)^{px}} &= (1 + pyG(y))^x = \sum_{n=0}^{\infty} \frac{x(x-1) \cdot \dots \cdot (x-n+1)}{n!} \cdot p^n \cdot y^n \cdot G(y)^n \end{aligned}$$

which is evidently in $1 + px\mathbb{Z}_p[[x, y]] + py\mathbb{Z}_p[[x, y]]$, thus by the lemma $F \in 1 + x\mathbb{Z}_p[[x, y]] + y\mathbb{Z}_p[[x, y]]$ $\square$

References

- [Kob96] N. Koblitz. *p-adic Numbers, p-adic Analysis, and Zeta-Functions*. Graduate Texts in Mathematics. Springer New York, 1996. ISBN: 9780387960173. URL: https://perso.univ-rennes1.fr/serge.cantat/Documents/Koblitz_ENS_2020.pdf.